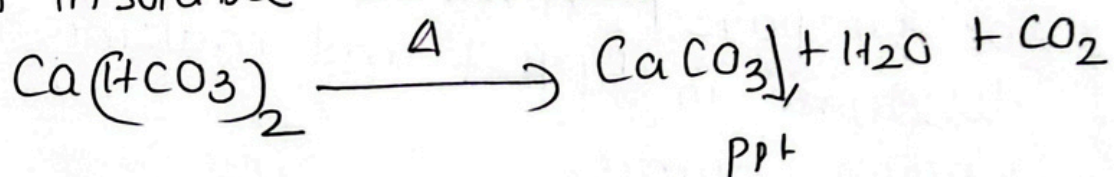


Chapter - 4

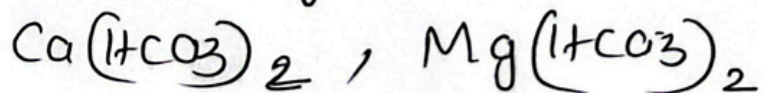
Environmental Chemistry

Types of Hardness

- (1) Temporary Hardness: It is due to the presence of dissolved bicarbonates of Ca and Mg. This is also known as carbonate hardness. Temporary hardness can be removed by boiling. On boiling soluble-bicarbonates are decomposed to form insoluble carbonates.



eg: of Temporary Hardness producing substances



- (2) Permanent or Non-carbonate Hardness:

It is due to presence of chlorides and sulphates of Ca and Mg. eg: CaCl_2 , CaSO_4 , MgCl and MgSO_4 . This type of hardness cannot be removed by boiling.

Degree of Hardness:-

Hardness is expressed in terms of calcium carbonate equivalent hardness.

It is the no: of milligrams of CaCO_3 present in one litre of water."

CaCO_3 is chosen as the standard for expressing hardness due to the following reasons.

- (1) CaCO_3 is the most insoluble salt that can be precipitated during water treatment.
- (2) Molecular mass of CaCO_3 is 100 & easy for calculation
- (3) A standard solution of hard water can be prepared from fixed amount of CaCO_3 , dissolving in dil. HCl and making up to the known volume.

Disadvantages of Hard water:

- (1) Hard water does not form ready lather with soap. Instead it produces sticky precipitates. This results in wastage of soap.
- (2) Hard water if used in boilers cause many troubles such as
 - (a) Scale and Sludge Formation:
 - " Scales are hard precipitates adhering at the inner wall of the boiler. Sludges are the loose and slimy precipitates.
 - (b) Priming and Foaming in boilers.

- (2) It causes boiler corrosion
- (3) Hard water can not be used in dyeing industry because it can not produce exact shades of colour.
- (4) Hard water if used for concrete mixing affects the hydration of cement and final strength of the concrete.
- (5) Hard water is not used in sugar industry due to formation of deliquescent sugar.
- Water Softening Methods :

[External Treatment Methods]

Ion-Exchange Resins:- They are insoluble cross-linked organic polymers with a porous structure and the functional groups attached to the chains are responsible for the ion-exchange properties.

cation exchange resins $[R-H^+]$

contain acidic functional groups like SO_3H , $COOH$ etc and are able to exchange their H^+ ions with Ca^{2+} and Mg^{2+} in the water sample.

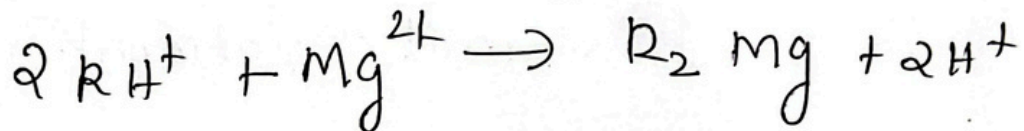
The anion Exchange Resin $[R-OH^-]$ contain functional groups like OH^- , amino groups etc. They become capable of

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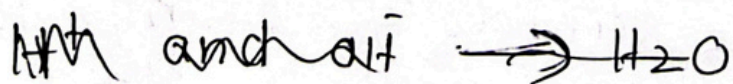
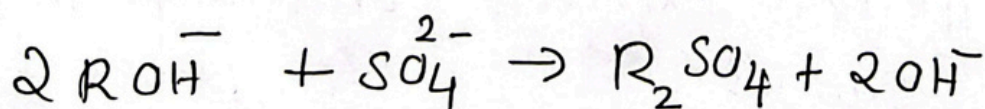
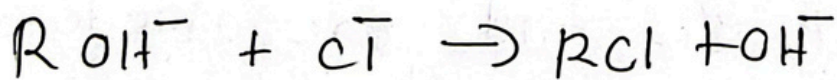
exchanging their OH^- ions with the anions present in water.

Process

Hard water is first passed through cation exchange resin, so that all the ~~cat~~ cations present in hard water (Ca^{2+} , Mg^{2+} etc) are exchanged with H^+ ions from the resin

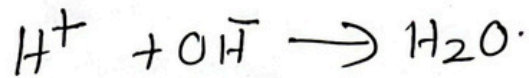


After this process hard water is passed through anion exchange column, so that all the anions (Cl^- , SO_4^{2-} etc) present in the water are removed and an equivalent amount of OH^- ions are released to the water from the column.



H^+ and OH^- ions liberated from the cations and anion exchangers

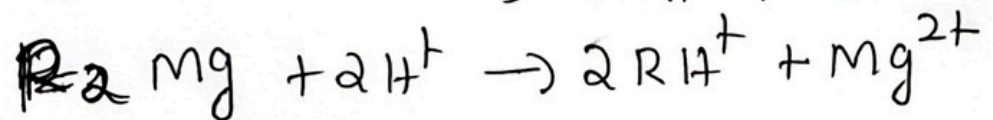
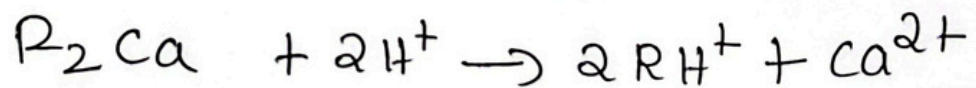
respectively, combine together to produce water molecule



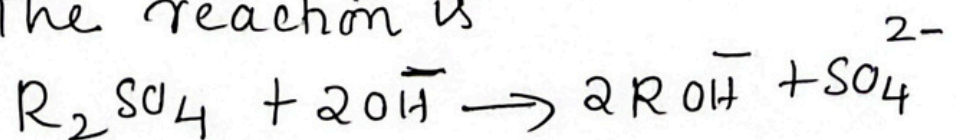
Thus the water coming out of the exchanger will be free of cations and anions. Hence the water is called deionised water or deionised water.

Regeneration

The exhausted cation exchange resin is regenerated by passing a solution of dil HCl or H_2SO_4 . The reaction is

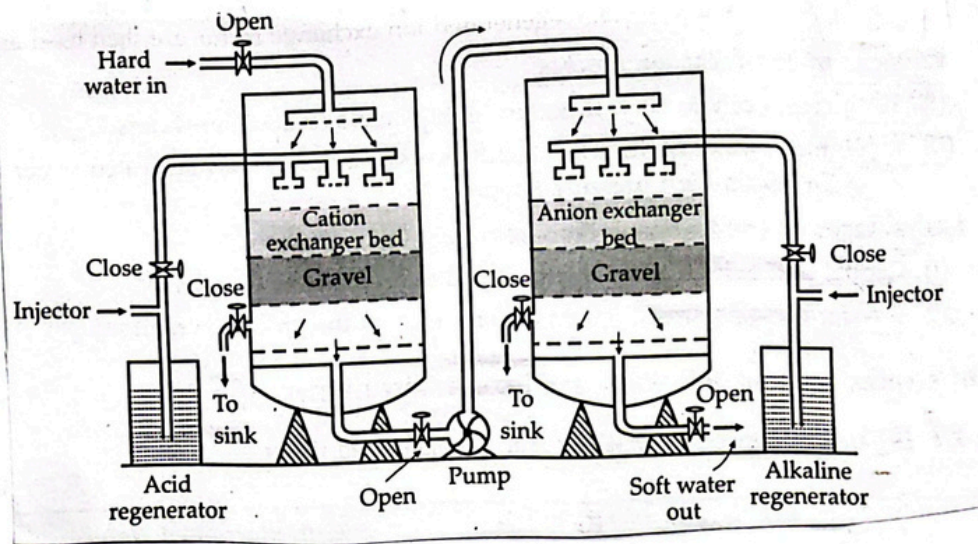


The exhausted anion exchange resin is regenerated by passing a solution of dil NaOH. The reaction is



Advantages

- It can be used to soften even acidic or alkaline waters.
- No sludge is formed
- The resin can be regenerated and reused.



Ion Exchange Process.

Desalination of Brackish water (Reverse Osmosis)

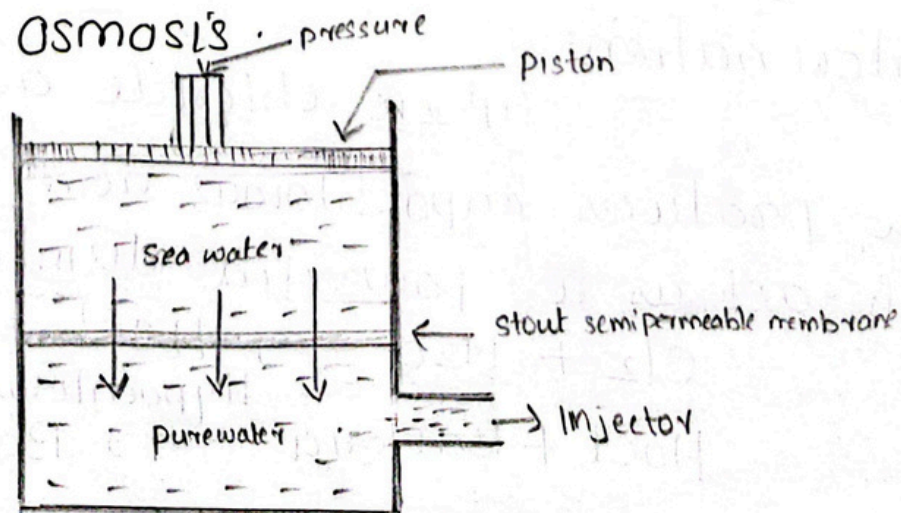
The process of removal of salt from (NaCl)

Saline water is known as Desalination.

Reverse osmosis is the most commonly used method for the desalination of water.

Reverse Osmosis

When two solutions of unequal concentrations are separated by a semi-permeable membrane, flow of solvent takes place from dilute to concentrated side due to osmosis. However if a pressure greater than osmotic pressure is applied on the concentrated side, reverse process takes place. i.e. solvent forced to move from concentrated side to dilute side across the semi permeable membrane. This is known as reverse osmosis.



Advantages

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- It is used to remove ionic as well as non-ionic, colloidal and organic matter.
- Low maintenance cost
- Low capital cost, Simplicity, low operating cost and high reliability.
- The life time of semi-permeable membrane is 2 years and membrane can be replaced within few minutes.

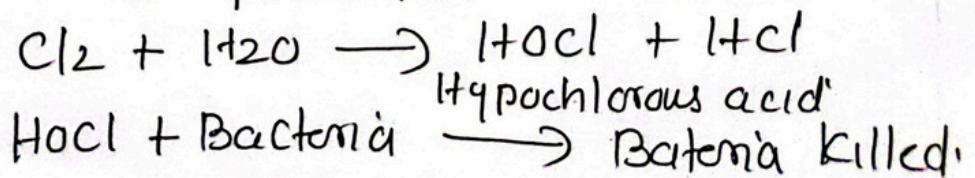
Disinfection

The process of killing the pathogenic bacteria and other micro-organisms from the ~~mal~~ water ^{to} and make it safe for use is called disinfection.

Disinfection can be done by the following methods.

(1) Chlorination

When chlorine added to water, produces hypochlorous acid (HOCl) which can act as a powerful disinfectant.

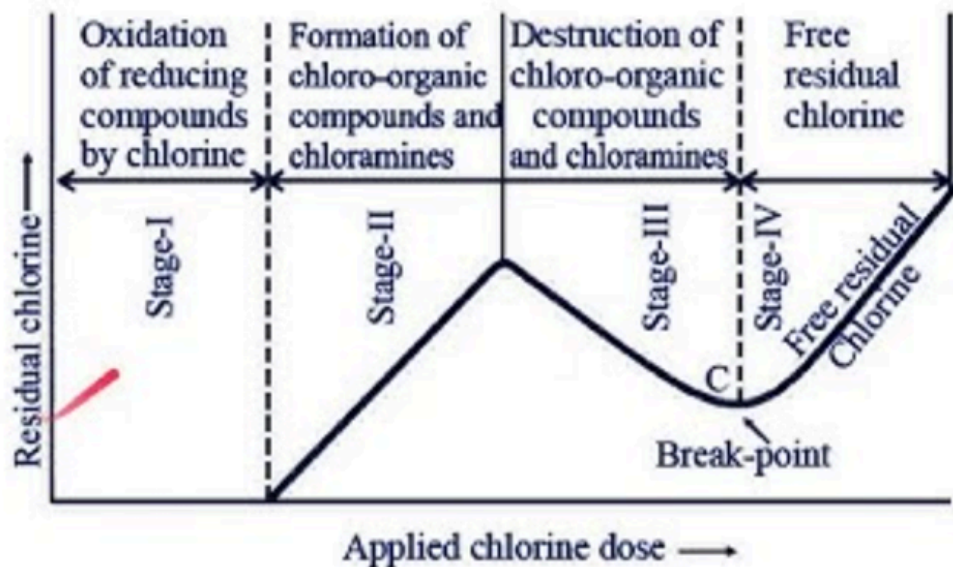


Advantages of chlorine

- (1) It is economical and effective
- (2) Storage is easy
- (3) can be used at low and high temperature.

Break-point chlorination ::

Break point chlorination involves addition of sufficient amount of chlorine to oxidise (a) organic matter (b) reducing substances (c) free ammonia in water, leaving behind mainly free chlorine. This free chlorine is used for disinfection process. The addition of chlorine at the dip or break is called break point chlorination. This indicates the point at which free residual chlorine begins to appear.



Advantages of break-point chlorination.

- (1) It completely oxidises the organic compounds ammonia and other reducing compounds.
- (2) It removes colour, odour and taste from water
- (3) It destroys all the disease-producing bacteria
- (4) It prevents the growth of any weeds in water.

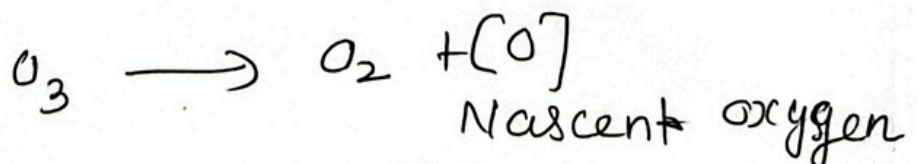
Q) Using Ozone

Ozone is an excellent disinfectant and it is used in the gaseous form.

O₃ is prepared by the reaction

$$3 \text{O}_2 \xrightarrow[\text{discharge}]{\text{Silent electric-}} 2 \text{O}_3$$

The O₃ thus produced is unstable and dissociates into O₂ and [O]



Advantages

- ~~Expensive method~~
- It simultaneously removes colour, odour, ~~to~~ and taste without leaving residue
- used for disinfection of mineral water

Disadvantages

- Expensive method
- can not be used in municipal water supply.

Dissolved oxygen

Dissolved oxygen refers to the level of free, non-compound oxygen present in water. It is an important parameter in determining the water quality. A low dissolved oxygen level can harm aquatic life.

Dissolved oxygen is the presence of free O_2 molecules in water. Dissolved oxygen levels fluctuate seasonally and vary with temperature and altitude (pressure).

Factor affecting DO

- (1) Temperature: As the temp increases DO decreases.:- Pure water at $30^{\circ}C$ can hold 7.8 ppm of dissolved oxygen but at $20^{\circ}C$ can hold up to 9.2 ppm. i.e. cold water hold more dissolved oxygen than warm water.
- (2) Altitude: Water hold less dissolved oxygen at higher altitudes due to lowering of atmospheric pressure.
- (3) Presence of oxidisable pollutants in water decreases the dissolved oxygen in water.

Biological Oxygen Demand [BOD]

BOD is defined as the amount of oxygen required by aerobic micro-organism for oxidation of biologically oxidisable matter present in one litre of sewage water for a period of 5 days at 20°C .

Significance of BOD

BOD is proportional to the amount of organic waste to be degraded aerobically. When BOD is high, Dissolved oxygen becomes low. Hence, greater the BOD, greater is the pollution. BOD is an indication of extent of pollution.

Chemical Oxygen Demand [COD]

COD is defined as the amount of oxygen required for the complete oxidation of biologically active and biologically inert materials present in one litre sewage water using chemical oxidising agents such as $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic conditions.

Significance of COD

COD is another Index of pollution

Due to rapid determination, COD is very important parameter in management and design of the treatment plants. It is used for calculating the efficiency of treatment plants.

Differences between Aerobic and Anaerobic oxidation

<u>Aerobic oxidation</u>	<u>Anaerobic oxidation</u>
• Take place in the presence of oxygen. • Oxidation by aerobic bacteria • products are non-offensive smelling • No biogas fuel is produced	• Take place in the insufficient amount of oxygen. • oxidation by anaerobic bacteria • products are offensive smelling • Bio-gas fuel is produced